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Control of ion content and nitrogen species using a mixed chemistry plasma for GaN grown at extremely high growth rates $>9 \mu\text{m/h}$ by plasma-assisted molecular beam epitaxy

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Utilizing a modified nitrogen plasma source, plasma assisted molecular beam epitaxy (PAMBE) has been used to achieve higher growth rates in GaN. A higher conductance aperture plate, combined with higher nitrogen flow and added pumping capacity, resulted in dramatically increased growth rates up to $8.4 \mu\text{m/h}$ using 34 sccm of N_2 while still maintaining acceptably low operating pressure. It was further discovered that argon could be added to the plasma gas to enhance growth rates up to $9.8 \mu\text{m/h}$, which was achieved using 20 sccm of N_2 and 7.7 sccm Ar flows at 600 W radio frequency power, for which the standard deviation of thickness was just 2% over a full 2 in. diameter wafer. A remote Langmuir style probe employing the flux gauge was used to indirectly measure the relative ion content in the plasma. The use of argon dilution at low plasma pressures resulted in a dramatic reduction of the plasma ion current by more than half, while high plasma pressures suppressed ion content regardless of plasma gas chemistry. Moreover, different trends are apparent for the molecular and atomic nitrogen species generated by varying pressure and nitrogen composition in the plasma. Argon dilution resulted in nearly an order of magnitude achievable growth rate range from $1 \mu\text{m/h}$ to nearly $10 \mu\text{m/h}$. Even for films grown at more than $6 \mu\text{m/h}$, the surface morphology remained smooth showing clear atomic steps with root mean square roughness less than 1 nm. Due to the low vapor pressure of Si, Ge was explored as an alternative n-type dopant for high growth rate applications. Electron concentrations from 2.2×10^{16} to $3.8 \times 10^{19} \text{ cm}^{-3}$ were achieved in GaN using Ge doping, and unintentionally doped GaN films exhibited low background electron concentrations of just $1\text{--}2 \times 10^{15} \text{ cm}^{-3}$. The highest growth rates resulted in macroscopic surface features due to Ga cell spitting, which is an engineering challenge still to be addressed. Nonetheless, the dramatically enhanced growth rates demonstrate great promise for the future of III-nitride devices grown by PAMBE. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4933278>]

I. INTRODUCTION

Group III-nitride semiconductors have seen enormous commercial growth in recent years for solid state lighting and power electronics. While the overwhelming majority of light emitting diodes (LEDs) are grown by metalorganic chemical vapor deposition (MOCVD), molecular beam epitaxy (MBE) has seen success in the power electronics market for the growth of high electron mobility transistors (HEMTs). Plasma-assisted MBE (PAMBE) typically uses a radio frequency (RF) plasma to generate reactive nitrogen species from inert nitrogen gas. PAMBE using a RF plasma has been shown to result in higher growth rates and improved surface morphology compared to other plasma techniques owing to lower ion content and reduced surface damage during growth.^{1–3} Some PAMBE reports have shown growth rates for GaN higher than $1 \mu\text{m/h}$,^{4,5} and a very recent publication has reported growth rates as high as $2.65 \mu\text{m/h}$.⁶

Nonetheless, PAMBE is typically performed at growth rates on the order of 0.1 to $1 \mu\text{m/h}$,^{7,8} substantially lower than the growth rates typically used in MOCVD growth of GaN, which commonly exceed $1\text{--}3 \mu\text{m/h}$.^{9–12} These relatively low growth rates limit the applicability of PAMBE for many device structures that require thick buffer layers to reduce defect densities resulting from heteroepitaxy. Although the ultra clean environment of MBE can be beneficial for the thick, undoped drift regions of p-i-n rectifiers, the slow growth rates still preclude the use of MBE for such devices. Finally, if III-nitrides, with their tunable band gap and strong light absorption,¹³ are ever to become viable for solar applications, thick indium-bearing layers must be demonstrated in order to reduce defect densities. Such layers with high indium mole fractions are impractical in MOCVD due to the low temperature requirements for growing indium bearing alloys, while high temperatures are required for cracking ammonia. The result for most MOCVD growth of InGaIn is very low growth rates and highly inefficient precursor usage.^{14,15} Contrarily, these high indium composition alloys are dramatically more suitable for PAMBE where substrate temperatures can be reduced in order to facilitate

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proper indium incorporation and no ammonia cracking is required. If rapid growth of indium-bearing layers can be achieved, then the potential of III-nitrides for photovoltaics improves dramatically.

In this study, we report on extremely high growth rates in PAMBE growth of GaN while maintaining a smooth surface microstructure. With added pumping capacity and a minor modification to the plasma source, a broad range of growth rates can be achieved, while a mixed chemistry plasma shows a further enhancement of the growth rate. Suppression of ion content is achieved by higher plasma pressure and through argon dilution. The maximum achieved growth rate of $9.8\ \mu\text{m/h}$ represents more than an order of magnitude increase compared to traditional growth rates in MBE and even exceeds those commonly used in MOCVD. To overcome a challenge introduced by the higher growth rates, we explore the use of alternative n-type doping methods. The high growth rates, resulting low unintentional doping levels, and excellent uniformity over large areas open the door for the use of MBE for novel devices utilizing thick indium-bearing layers, as well as existing devices where the traditional slow growth in MBE was previously not viable.

II. EXPERIMENTAL

All films in this study, unless otherwise specified, were grown in a Riber 32 MBE system with a base pressure of approximately 8×10^{-11} Torr. The substrates were single-side polished 1×1 cm sapphire, 1×1 cm GaN templates, or 2 in. sapphire wafers. All substrates were sputter coated with $2\ \mu\text{m}$ of tantalum on the back side for heat absorption and spreading. After cleaning in a 4:1 solution of $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$, the substrates were loaded into an introductory chamber on an indium-free molybdenum block and then outgassed at 150°C for 20 min. The samples were then transferred to the growth chamber and were outgassed for an additional 10 min at 600°C for the GaN template substrates or 850°C for the sapphire substrates. Any substrate temperatures listed herein were those measured by thermocouple. For all film growths, nitrogen was supplied by a Veeco UNI-Bulb RF plasma source, with several modifications, which will be described later. The gas flow and RF power were varied throughout the study to achieve the desired surface conditions and also to explore their effects on growth rate. Aluminum was supplied by a cold-lip effusion cell, gallium was supplied by a standard effusion cell and/or a Veeco dual-filament SUMO cell, and germanium was supplied by a standard effusion cell for n-type doping. The total Ga fluxes used in this study varied from $\sim 8 \times 10^{-7}$ Torr beam equivalent pressure (BEP) up to $\sim 3.6 \times 10^{-6}$ Torr BEP.

The growth techniques employed herein were a combination of traditional MBE and a shuttered variation of MBE called Metal-Modulated Epitaxy (MME), the details of which can be found elsewhere.^{16–19} For the sapphire substrates, a low-temperature nitridation was performed for 1 h at 200°C ,^{20,21} followed by a 10 nm AlN nucleation layer under N-rich conditions (III/V ≈ 0.7) at 700°C . Then, a thicker AlN buffer layer was grown using MME at 850°C under Al-rich conditions (III/V ≈ 1.5), while the substrate

temperatures during GaN growth were typically 600 to 650°C . This MME technique has been shown to achieve extremely high p-type doping,^{22,23} control phase separation in InGaN,^{24,25} and result in excellent uniformity and material utilization due to the temperatures substantially lower than traditional PAMBE.

The film growth was monitored *in situ* by reflection high energy electron diffraction (RHEED), and the plasma emission spectra were monitored through the rear window of the plasma source looking through the pyrolytic boron nitride (PBN) bulb using an Ocean Optics HR-2000+ES spectrometer in the wavelength range of 200–1100 nm. Post-growth structural analysis was performed by x-ray diffraction (XRD) using a Philips X'Pert Pro MRD, and the surface morphology of the films was characterized by a Veeco Dimension 3100 atomic force microscope (AFM) operated in tapping mode. Film thicknesses were verified by either contact profilometry or spectroscopic ellipsometry. Hall Effect was employed to determine the electrical properties of select films using a 0.3 T fixed magnet or a custom Hall effect system with a 1.5 T electro-magnet.

III. RESULTS AND DISCUSSION

A. Plasma and chamber modifications

In order to facilitate the dramatically increased growth rate in this study, several important modifications were made to the plasma source and the growth chamber. Two additional cryo pumps (CTI-10) were added, each on a separate 10 in. outer diameter conflat flange. The overall pumping system for the growth chamber consisted of one CTI-8 cryo pump, two CTI-10 cryo pumps all used during growth, and one ion pump and a titanium sublimation pump used when not growing. With the added cryo pumps, the pumping capacity during growth was more than tripled compared to the previous single cryo pump configuration. This increased capacity resulted in a substantial reduction in the chamber pressure during growth, never exceeding 10^{-4} Torr even with more than 30 standard cubic centimeters per minute (sccm) of N_2 flow. The lower chamber pressure reduces wear on effusion cell filaments and could serve to decrease scattering of the incident flux to improve growth rates.

In addition to the added pumping capacity, the plasma source itself was also slightly modified. The original plasma aperture was replaced with a custom $5.6\times$ higher conductance aperture in order to allow for higher gas flow while maintaining the correct plasma mode. With the original aperture, the plasma switched to a dim mode resulting from excessive back pressure behind the aperture with a nitrogen flow of just 3.25 sccm, but remained lit in the bright mode with flow less than 1 sccm. Using the new high-conductance aperture, the plasma required substantially higher nitrogen flow of at least 3 sccm to remain ignited, but continued in the bright mode up to the maximum tested nitrogen flow of 34 sccm.

B. Growth rate

In order to initially characterize the plasma source, the GaN growth rate was determined using post-growth

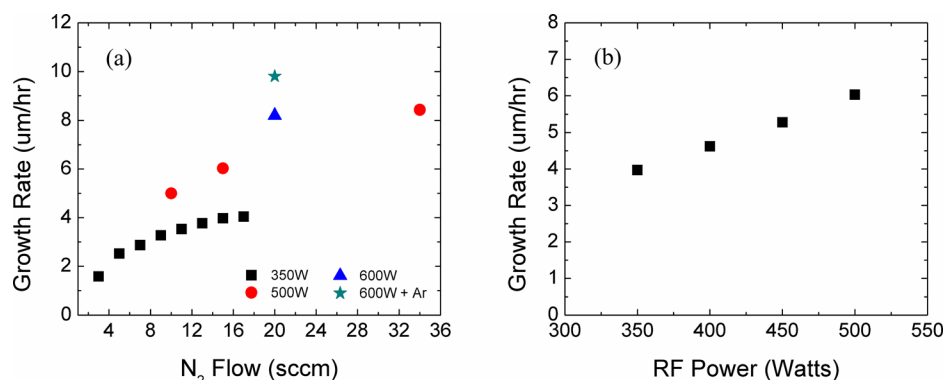


FIG. 1. (a) GaN growth rate versus nitrogen flow for various RF power conditions and (b) growth rate vs. RF power for a fixed nitrogen flow of 15 sccm.

thickness measurements combined with *in situ* estimation of III/V flux ratios by transient RHEED analysis. The RHEED and post growth methods matched to less than 5% accuracy in all cases. To avoid desorption of the Ga flux, the samples used for growth rate determination were grown at a low substrate temperature of 600 °C under metal-rich conditions. The nitrogen flow was initially varied from 3 sccm to 17 sccm with a fixed power of 350 W, and then the plasma power was varied from 350 W to 500 W with a fixed nitrogen flow of 15 sccm. This resulted in growth rates of 1.5 μm/h to 6.1 μm/h. Using pure nitrogen at 500 W RF power and increasing the flow to 34 sccm using a second mass flow controller yielded a growth rate of 8.4 μm/h.

To assist in the initial striking of the plasma after baking the system, a light flow of argon gas was added, which is easier to ignite than pure nitrogen. After the plasma ignited, the argon flow was cut off and the plasma transitioned to pure nitrogen. However, it was observed that this procedure resulted in an increase in the active nitrogen flux for films grown immediately after the use of argon. As such, another test was performed to determine the effect of argon on the growth rate for various conditions. From this test, it was discovered that the growth rate increased by approximately 20% to 50% at high and low nitrogen flows, respectively, with the addition of 7.7 sccm argon into the plasma. Shown in Figure 1 is the overall growth rate as a function of these various plasma conditions, with growth rates from approximately 1.5 μm/h to 9.8 μm/h. Moreover, the uniformity for these high growth rates was excellent as measured across a 2 in. wafer by spectroscopic ellipsometry, demonstrating a standard deviation of just 2% for a 4 μm thick film.

The maximum growth rate was achieved using 600 W RF power with 20 sccm nitrogen and 7.7 sccm argon flows, which is limited by our current mass flow controller hardware configuration. This condition resulted in an extremely high growth rate of 9.8 μm/h, which is, to our knowledge, the highest ever reported for GaN grown by MBE. Under this same condition using 20 sccm nitrogen without argon, the growth rate dropped to 8.2 μm/h.

While the demonstration of high growth rates is beneficial for future applications, the films must also remain sufficiently high quality in order to be useful. Shown in Figure 2 is a representative RHEED image during growth, as well as a 2 × 2 μm AFM image for a film grown at 6.1 μm/h. As shown in Figure 2(a), during the growth of these films under

metal-rich conditions at 600 °C, the RHEED patterns remained streaky and a 2 × 2 reconstruction was observed, indicating a smooth growth surface. The AFM image in Figure 2(b) shows excellent dislocation-mediated step-flow morphology with clear atomic steps and spiral hillocks and a root mean square (RMS) roughness of just 0.8 nm. Such a surface structure is typical for high quality GaN grown by MBE and is also seen at much lower growth rates.^{8,26} For comparison, a similar AFM surface morphology is shown in Figure 2(c), which also demonstrates the same smooth step-flow growth at a lower growth rate of ~2 μm/h. This observation confirms that the 2D growth mode is preserved even at growth rates more than an order of magnitude higher than are commonly used in GaN MBE.

In an effort to understand the physical explanation for the growth rate change with versus without argon, the plasma

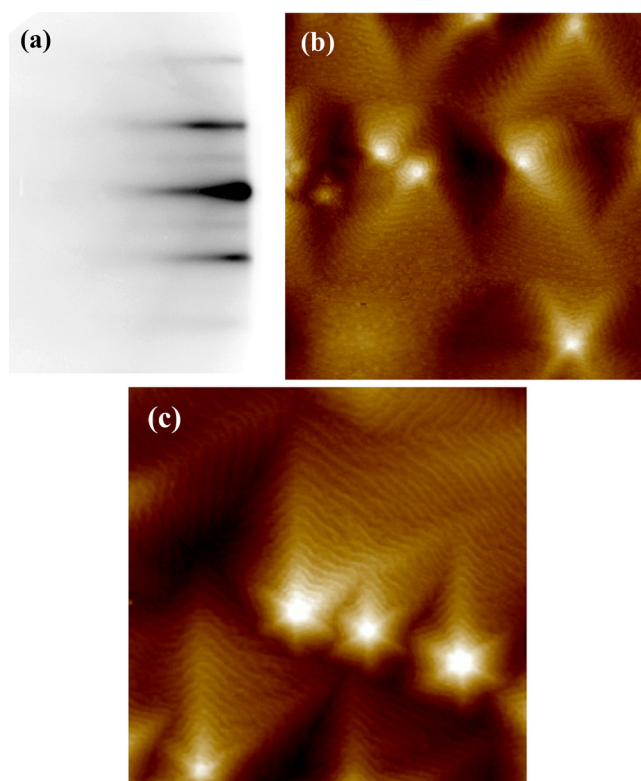


FIG. 2. (a) RHEED image taken after growth, (b) resulting AFM scan for GaN film grown at 6.1 μm/h at a substrate temperature of 600 °C, and (c) additional AFM scan for a lower growth rate of 2.0 μm/h (height scale = 7 nm, RMS roughness = (b) 0.8 nm, and (c) 1.0 nm).

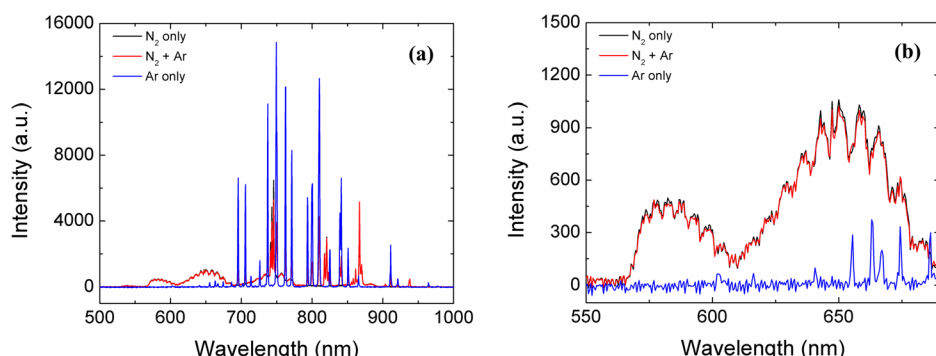


FIG. 3. (a) Full range plasma spectra from 500 to 1000 nm and (b) close-up of the 550–690 nm wavelength range (right) associated with excited molecular nitrogen species. Plasma power of 500 W using gas flows: 20 sccm N₂ only (black), 20 sccm N₂ + 7.7 sccm Ar (red), and 7.7 sccm Ar only (blue).

emission spectra were compared using a simple fiber-coupled spectrometer on the rear plasma window and are shown in Figure 3. The spectra in Figure 3 represent 20 sccm pure N₂ (black), a mixed plasma using 20 sccm N₂ with 7.7 sccm Ar (red), and finally a pure 7.7 sccm Ar (blue) using 350 W of RF power in all cases. Figure 3(b) shows the 550 nm to 690 nm region corresponding to excited molecular nitrogen (N₂^{*}), which has previously been shown to directly affect growth rate of III-nitrides in plasma MBE.^{4,5,27}

From Figure 3(b), it is clear that the addition of argon had no meaningful effect on the excited molecular nitrogen bands, in contrast to other findings using a mixed N₂/Ar plasma.²⁸ However, the intensities of the argon-related emission lines were dramatically reduced by a factor of 3–4× in the mixed plasma compared to the pure argon plasma. Thus, it is possible that the reduction in argon-related emission intensity in the mixed plasma could be due to the transfer of kinetic energy from the argon species to the nitrogen species, enhancing the formation of other active nitrogen species besides molecular species and increasing the growth rate.

To further explore the effects of argon in the plasma, the manipulator was put into the flux measurement position while measuring the collector current using a Keithley 6517B electrometer with the collector bias held at 0 V. During the measurement, the grid voltage was set to 0 V and the flux gauge filament was turned off with the collector shielded by earth ground except for a small opening facing the plasma source. This ensured that the only ionized/charged species reaching the collector were those coming from the plasma source itself and/or species created near the collector of the ion gauge. As such, the collector functioned similar to a Langmuir probe where the collector current is related, albeit indirectly, to the ion content of the plasma as will be discussed below.^{29,30} Shown in Figure 4 is the resulting probe current from this experiment at 300 W RF power as a function of the nitrogen/argon concentrations as well as the approximate pressure inside the plasma bulb (i.e., behind the aperture plate) as calculated by the mass flow rate and the theoretical conductance of the aperture plate.

It should be first emphasized that this experimental configuration with zero bias will result in the collection of both electrons and ions and is thus related to but not equal to the ion content. With a positive bias applied to the grid, positively charged primary ions should be rejected and negatively charged primary electrons attracted, or vice versa for a negative bias on the grid. However, in this experiment, it

was found that the probe current was always negative with a magnitude of at least 10 nA regardless of applied grid bias up to ± 20 V. This was not a measurement offset error as the probe current was less than 10 pA, while the plasma was turned off. Thus, the only reasonable explanation for this persistent large negative current is the presence of a high density of secondary electrons created by the primary high energy species (ions and/or electrons from the plasma), which overwhelm the primary ion and electron currents from the plasma itself. Secondary electrons can be generated from nearby surfaces via impact ionization or photoemission, the latter having been acknowledged as a complicating factor but discounted herein as has been done in prior more detailed characterization efforts.²⁹ Because the trends in optical intensity (described later) do not follow the trends in probe current, the role of photoemission is assumed negligible in the present case. However, both of these mechanisms (impact ionization and photoemission) suggest that the plasma produces sufficiently high energy primary species to cause ionization and, thus, potential ionizing damage to the sample during growth. Thus, we do not treat the measured collector currents as a direct measure of ion current but instead an impact ionization enhanced current related to but not equal to ion current. Therefore, regardless of sign of the current, variations in the current are assumed to originate from changes in the high energy, potentially damaging, primary plasma species.

From Figure 4, it is clear that, in the lower pressure regime, the pure nitrogen plasma results in a dramatic increase

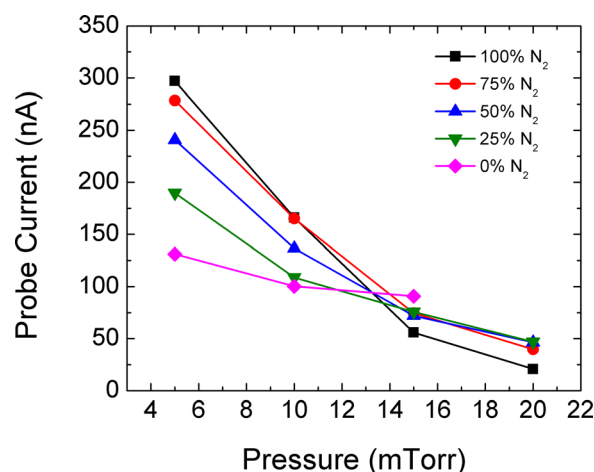


FIG. 4. Probe current as a function of pressure measured for various plasma compositions.

of more than 100% in probe current compared to the pure argon plasma. Additionally, with proper dilution of the nitrogen with argon at lower pressure/flow, the probe current, and thus inferred ion/electron primary flux, can be dramatically reduced in order to minimize any detrimental effect of high energy species bombarding the sample surface and causing damage. Higher plasma pressure suppresses the ion content, making the ion content relatively unaffected by plasma composition. This observation of probe current shows two mechanisms to control plasma damage due to high energy species: in the lower flow regime by argon dilution and in the higher flow regime via pressure.

Shown in Figures 5(a) and 5(b) are the emission intensities at 300 W RF power for the 650 nm and 821 nm plasma emission lines relating to molecular nitrogen and atomic nitrogen transitions, respectively. The 650 nm emission from excited molecular transitions in Figure 5(a) shows a monotonic increase in intensity with increasing nitrogen composition in the plasma, and also with increasing pressure. Interestingly, a substantially different trend can be seen in the 821 nm atomic emission in Figure 5(b) where the intensity trends do not follow the 650 nm molecular increases. At lower nitrogen compositions of 25% and 50%, the atomic nitrogen intensity increases with increasing pressure. However, above 50% nitrogen, there is a peak in the atomic emission intensity at ~ 10 mTorr, which then decreases at higher pressure. In the case of a pure nitrogen plasma, the atomic emission intensity is nearly 55% higher at 10 mTorr compared to 20 mTorr. The overall atomic nitrogen intensity is highest for a diluted plasma with just 50% nitrogen at 20 mTorr.

These trends in the atomic nitrogen intensity can further explain the increased growth rate with argon as described previously and as shown in Figure 1(a). In the literature, there is no clear consensus which nitrogen species (atomic or molecular) is the primary contributor to growth during PAMBE of III-nitrides.^{4,5,31} In this work, at the higher plasma pressures where the growth rate is highest, there is a substantial shift in the trends of atomic nitrogen intensity compared to molecular nitrogen. In the high pressure/flow case, the dilution with $\sim 25\%$ – 50% argon minimally reduces the molecular emission but substantially increases the atomic emission intensity of the plasma. Thus, one would assume that the 20%–25% increase in growth rate observed in Figure 1(a) results from an enhancement in atomic nitrogen, not the traditionally assumed excited molecular species. At

lower plasma chamber pressures, 10 mTorr or less, both the molecular and atomic nitrogen intensities are highest using a pure nitrogen plasma. But, at higher pressures like those used to achieve growth rates above $6 \mu\text{m/h}$, the atomic nitrogen is maximized for a mixed chemistry plasma with $\sim 50\%$ argon dilution. The maximum growth rate of $9.8 \mu\text{m/h}$ reported here was achieved at a pressure of ~ 30 mTorr with a nitrogen composition of $\sim 72\%$. A more in-depth study is currently underway in order to explore the relative effects of atomic and molecular nitrogen and the various plasma conditions on growth rate and structural/optical quality. But, from this preliminary characterization of the plasma, it is clear that both the atomic and molecular nitrogen species are important to consider, especially at the extremely high growth rates reported herein.

In this study, the maximum growth rate achieved was not limited by the plasma source, but rather by the 20 sccm nitrogen mass flow controller, which indicates that even higher growth rates should be possible. In addition, the growth rate can be reduced by varying the nitrogen and argon at low flow rates. With pure nitrogen, the plasma extinguished at a flow of ~ 3 sccm due to insufficient back-pressure. However, it could be sustained by using a combination of 1 sccm of nitrogen with 7.7 sccm of argon, resulting in more traditional growth rates closer to $1 \mu\text{m/h}$. By lowering the argon content further, an additional reduction in growth rate might also be achievable to extend the range of possible growth rates. The varied growth rate from $\sim 1 \mu\text{m/h}$ up to $9.8 \mu\text{m/h}$ represents nearly an order of magnitude achievable range of growth rates, which can be tuned for rapid growth of thick buffer layers or precisely controlled quantum wells.

C. Doping at high growth rates

In order to achieve sufficient n-type doping at these elevated growth rates, some reconsideration of the Si doping method is needed. Typically, n-type doping in MOCVD growth using Si is limited to the low or mid 10^{19}cm^{-3} range due to tensile stress and film cracking during cool down, as well as morphological degradation at higher Si concentrations.^{32,33} In MBE, due primarily to the lower substrate temperatures, this is less of an issue.³⁴ Using MME, we have previously achieved extremely high Si doping with an electron concentration up to $2.5 \times 10^{20} \text{cm}^{-3}$ and resistivity of $5 \times 10^{-4} \Omega \text{cm}$ for a 900 nm thick film with RMS roughness

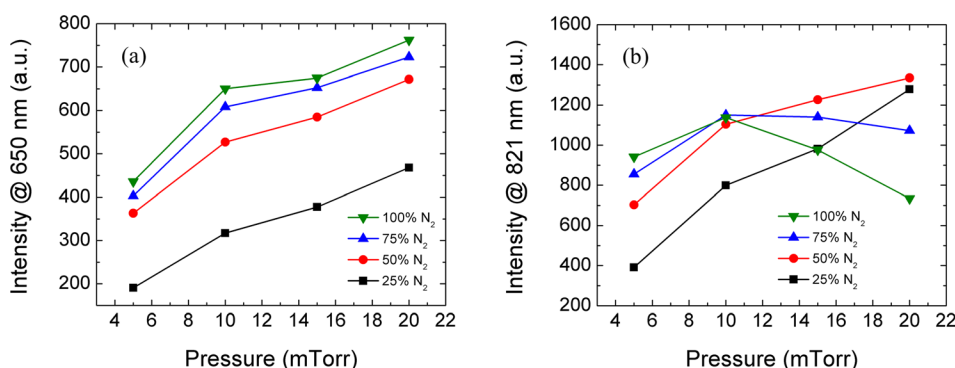


FIG. 5. (a) Plasma emission intensity for various plasma pressures and compositions at 650 nm related to molecular nitrogen and (b) 821 nm (right) related to atomic nitrogen.

of just 0.3 nm. This film was grown at $\sim 1 \mu\text{m/h}$, and the Si dopant cell required a temperature in excess of 1200°C . However, in order to achieve this high level of doping at elevated growth rates, an even higher cell temperature would be required, pushing the power limits of the cell and reducing effusion cell and/or crucible lifetime.

To circumvent this doping challenge at high growth rates, the use of Ge was explored as an alternative.^{35,36} Compared to Si, Ge has approximately two orders of magnitude higher vapor pressure for the same temperature.^{37,38} Or, to achieve the same vapor pressure as Si, a Ge cell could be run at approximately 200°C lower temperature. Ge has been shown to have similar activation energy to Si in GaN and induces substantially less stress on the crystal due to its closer atomic size match to the Ga atom it substitutes in the lattice.^{39,40} Given the lower temperature requirements with similar activation, Ge would seem an excellent alternative for high growth rate GaN. The use of Ge was investigated by varying the Ge cell temperature from 760°C to 1000°C with a fixed growth rate and Ga flux, as shown in Figure 6. Sapphire substrates were used for this study to provide complete electrical isolation and minimize the effect of surface contamination. A $\sim 500\text{nm}$ thick AlN buffer layer was grown by MME after low-temperature nitridation. The resulting AlN, grown at nearly $2 \mu\text{m/h}$, showed (002) rocking curve (omega scan) full width at half maximum (FWHM) of around 80 arcsec with RMS roughness of $\sim 0.5\text{ nm}$. For the Ge-doped GaN grown at 650°C , MME was employed using a 10 s open/10 s closed shutter modulation with a III/V ratio of approximately 1.8, and both Ge and Ga were shuttered simultaneously. The plasma condition for these films was fixed at 350 W RF power with 4 sccm N_2 flow, yielding a growth rate of approximately $2 \mu\text{m/h}$.

As shown in Figure 6, the electron concentration exhibits a highly predictable exponential dependence on the Ge cell temperature. The electron concentrations achieved varied from $2.2 \times 10^{16} \text{ cm}^{-3}$ at a Ge cell temperature of 760°C up to $3.8 \times 10^{19} \text{ cm}^{-3}$ at a Ge cell temperature of 999°C . These low dopant cell temperatures leave substantial headroom for high n-type doping at high growth rates while simultaneously reducing wear on the cell and crucible. As a

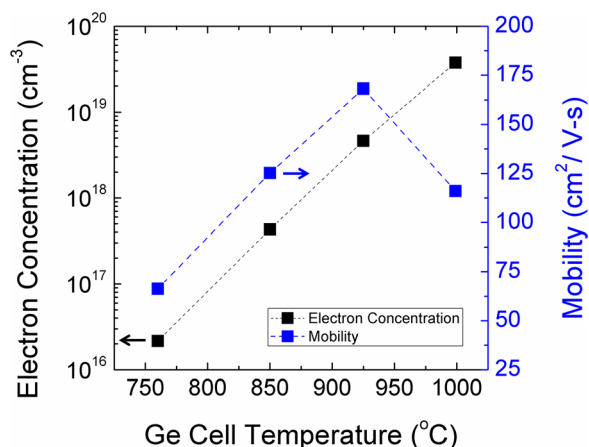


FIG. 6. Electron concentration (black) and mobility (blue) as a function of Ge cell temperature.

baseline, several unintentionally doped (UID) GaN films were grown by MME in order to characterize the background electron concentration. Since MBE grown films have background impurity concentrations inversely proportional to growth rate, low background impurity concentrations are expected at these high growth rates. Indeed, with the same conditions as discussed above, the UID GaN films exhibited repeatable background electron concentrations of $1\text{--}2 \times 10^{15} \text{ cm}^{-3}$ with resistivity on the order of $50\text{--}100 \Omega \text{ cm}$. The mobility, shown in blue in Figure 6, initially rises with increasing Ge cell temperature then begins to fall once the electron concentration exceeds $\sim 4 \times 10^{18} \text{ cm}^{-3}$. This trend, along with the relatively low overall mobility, can be explained in the context of relatively high dislocation density resulting from the immature nature of these initial high growth rate studies.

In these non-optimized $1.8 \mu\text{m}$ thick GaN films grown on AlN buffer layers on sapphire, the XRD GaN (002) and (102) omega rocking curve FWHM are typically on the order of 250 arcsec and 1500 arcsec, respectively. Based on simple estimations from these two rocking curves, the threading dislocation density in these Ge-doped films is expected to be on the order of 5×10^9 to $1 \times 10^{10} \text{ cm}^{-2}$.^{41–43} With these relatively high dislocation densities, the mobility is strongly affected by dislocation scattering at low carrier concentrations. At higher carrier concentrations, the dislocation scattering is minimized due to charge screening, while at very high doping concentrations, the mobility again decreases due to ionized impurity scattering.^{44,45} Thus, while the mobility in these Ge-doped samples is relatively low, it is well-described by physical models and limited by dislocation density.⁴⁶ It is expected that the mobility should be substantially higher when grown on lower dislocation density GaN templates or freestanding GaN.

D. Challenges at high growth rates

While the above results are promising for a number of possible applications, there are still challenges to be addressed at these extremely high growth rates. During the span of these studies, the most commonly encountered issue was macroscopic surface features on the film. Shown in Figure 7 is an optical microscope image for a film grown at $8.2 \mu\text{m/h}$.

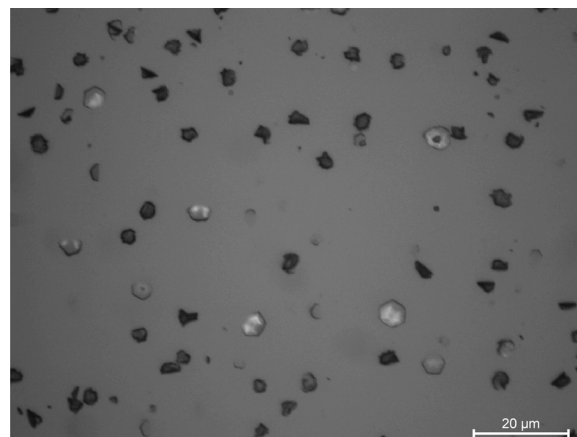


FIG. 7. Optical microscope image of GaN grown at $8.2 \mu\text{m/h}$.

8.2 $\mu\text{m/h}$, which exhibits dark clusters and transparent hexagonal pyramids. The most likely explanation for these surface features is “spitting” from the Ga cell and subsequent nitridation/non-epitaxial crystal formation. At very high fluxes, like those used here, Ga has a tendency to “spit,” wherein large clusters of Ga exit the end of the effusion cell rather than an ideal uniform flux. In Figure 7, the dark clusters are likely metallic gallium clusters that were unable to incorporate correctly or formed a nitride skin, while the transparent hexagonal pyramids are expected to be GaN crystallites.

Given the smooth morphology in the AFM image in Figure 2(b), the surface features appear to be associated with the Ga source(s) rather than a degradation of the growth mode at elevated growth rates. The primary dual-filament Ga cell in this work was designed to mitigate the spitting issue but, at these extremely high growth rates, that alone was insufficient to eliminate the surface features altogether. In our system, the simultaneous use of two Ga cells did show some reduction of the large surface features but similarly did not completely eliminate the problem. Thus, some improvements in cell or crucible design would be necessary for this high growth rate to become commercially viable. A valved cell, aperture plate, or dramatically tapered crucible could potentially overcome this issue by removing or substantially decreasing the direct line-of-sight from the Ga pool to the sample surface. At elevated growth rates above 4 $\mu\text{m/h}$, the onset of Ga spitting, and the resulting surface features, makes Hall effect measurements impractical. As such, no accurate doping or UID electron concentration observations for high growth rate films can be made until the spitting issue is resolved.

IV. CONCLUSION

In conclusion, we have demonstrated the use of a mixed chemistry plasma in order to achieve GaN growth rates of nearly 10 $\mu\text{m/h}$ by plasma assisted MBE. By adding argon to the plasma gas flow, substantial growth rate increases can be achieved while also allowing lower growth rates through proper dilution of the nitrogen at low flow rates, resulting in a growth rate range of $\sim 1 \mu\text{m/h}$ up to 9.8 $\mu\text{m/h}$. The Langmuir probe current, and thus indirectly the ion content of the plasma, was measured *in situ*. It was found that the high energy species hitting the substrate could be reduced by changing the plasma pressure and the argon/nitrogen chemistry, providing an additional mechanism to tune the growth rate while simultaneously minimizing possible surface damage. Further characterization of the plasma emission spectra points to a significant contribution from atomic nitrogen to the net growth rate, where the atomic nitrogen is maximized for a mixed argon/nitrogen plasma at higher plasma chamber pressures. The surface morphology of the high growth rate MME films shows no degradation of the growth mode, maintaining a dislocation-mediated step-flow growth with sub-nanometer RMS roughness even at relatively low substrate temperatures. In order to address doping challenges associated with Si doping at high growth rates, Ge was explored as a possible alternative. Electron concentrations were achieved from $2 \times 10^{16} \text{ cm}^{-3}$ up to $4 \times 10^{19} \text{ cm}^{-3}$ for growth rates of

$\sim 2 \mu\text{m/h}$ while operating the Ge dopant cell nearly 200 °C cooler than would be required for equivalent Si doping. Repeatable background electron concentrations of $1\text{--}2 \times 10^{15} \text{ cm}^{-3}$ were achieved for unintentionally doped films grown under these conditions. While these results are promising, films grown at higher growth rates exhibited large surface features, most likely due to Ga “spitting” from the effusion cell. Improved crucible or cell designs should minimize these detrimental surface features. Nonetheless, such an engineering challenge is relatively straightforward, and the preservation of an ideal step-flow growth mode proves the fundamental viability of these high growth rates.

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